

no bandwidth tuning

Energy Distance

- Definition: $\|\mu - \nu\|_{ED} = \left(\mathbb{E}\|X - Y\| - \frac{1}{2}\mathbb{E}\|X - X'\| - \frac{1}{2}\mathbb{E}\|Y - Y'\| \right)^{1/2}$
- It's an MMD with kernel $k_{ED}(x, y) = -d(x, y)^p$, $0 < p < 2$
- Scale-free, **no bandwidth tuning**.
- Popular in classical statistics (two-sample testing, distance correlation).

IPMs on Diracs



$$(\alpha, \beta) = (\delta_0, \delta_t)$$

